

LOW MOLECULAR WEIGHT NBR: PATENT LANDSCAPE ON CONTROLLED MOLECULAR-WEIGHT REDUCTION, SOLUTION POLYMERIZATION, AND POST-POLYMER DEGRADATION

1. ABSTRACT

This report synthesizes a set of patents relevant to low molecular weight nitrile butadiene rubber (NBR), with emphasis on routes that deliberately reduce molecular weight, lower viscosity, or improve processability. The primary target is low-MW NBR or NBR-derived materials with controlled chain length, typically achieved either during polymerization by solution/emulsion control and chain-transfer/modifier strategies, or after polymerization by olefin-metathesis degradation. The landscape is broader than classical NBR synthesis alone: several patents address low-viscosity or low-MW polymer architectures that are conceptually transferable to NBR processing, while others directly disclose NBR depolymerization or molecular-weight reduction.

The most directly relevant disclosures are the solution-phase NBR patent EP2478021B1, which uses free-radical polymerization of butadiene/acrylonitrile in organic solvent with a modifier (regulator) and demonstrates deliberate low-MW NBR preparation by increasing regulator loading to 2.22 phm, and US9346908B1, which reduces NBR molecular weight by olefin-metathesis using ruthenium catalysts on magnetic carriers, enabling catalyst recovery and reuse. EP2478021B1 also shows that solution polymerization can be used to prepare NBR terpolymers and RAFT-derived low-MW macromers, with polymerization at 100°C and 3 bar butadiene. US9346908B1 demonstrates that 6 wt% NBR/chlorobenzene solutions treated with 2 wt% 1-hexene and 5 wt% loaded catalyst at room temperature for 2 h can yield lower-MW NBR while preserving recoverability.

Other patents in the set are not NBR synthesis patents per se, but they contribute relevant process concepts: controlled solution polymerization with precise molecular-weight tuning (US10167348B2), associative/telechelic architectures for viscosity control (US11905379B2), living radical graft copolymerization with low-MW side chains (US9988480B2), and adiabatic bulk polymerization with chain-transfer control (US10465017B2). Together, these disclosures indicate that low-MW NBR can be approached through solvent-based radical polymerization with regulator/chain-transfer control, or through post-polymer metathesis degradation, with isolation by precipitation or magnetic separation and with strong emphasis on viscosity/processability

management.

2. METHODOLOGY

PRIMARY PATENT EVIDENCE

Patent 1

- Patent Number: EP2478021B1
- Patent Title: Nitrile rubbers and production of same in organic solvents
- Assignee: Not disclosed
- Jurisdiction: EP
- Publication Year: 2015
- Legal Status: Unknown
- Polymer Type: Not disclosed
- Relevance: Automatically extracted candidate

Polymerization / Synthesis Method - Synthesis method: This patent discloses solution polymerization of nitrile rubbers in organic solvent using free-radical initiation and a modifier/regulator to control molecular weight. The examples show preparation of NBR from acrylonitrile and 1,3-butadiene in DMAc, with oxygen removal by argon purge, pressurized butadiene addition, thermal initiation at 100°C, and isolation by ethanol precipitation. A deliberately low-molecular-weight NBR is prepared by increasing regulator concentration. - Initiator Vazo® 88: 284.9 mg (1.17 mmol, 0.31 phm) — Example 1 - DoPAT: 204 mg (0.583 mmol, 0.22 phm) — Example 1 - Solvent: 50 mL dimethylacetamide (51 phm) — Example 1 - Acrylonitrile: 43 mL (653.3 mmol, 37 phm) — Example 1 - 1,3-Butadiene: 90 mL (1077 mmol, 63 phm) — Example 1 - Butadiene pressure: 3 bar — Example 1 - Polymerization temperature: 100°C — Example 1 - Polymerization time: 9 h — Example 1 - Final conversion: 39% — Example 6 - Regulator amount: 2.22 phm — Example 7 - Polymerization time: 9 h — Example 7 - Final conversion: 48% — Example 7 - Mn: 12,000 g/mol — Example 8 - Styrene: 2.79 g (26.8 mmol, 50 phm) — Example 8 - Polymerization temperature: 100°C — Example 8 - Polymerization time: 13 h — Example 8 - Final conversion: 9.2% — Example 8 - Vazo® 88 post-dose: 128 mg (0.523 mmol, 0.15 phm) — Examples 18, 20, 23 - Post-dose solvent: 5 mL dimethylacetamide — Examples 18, 20, 23 - Butadiene pressure: 3 bar — Examples 17-23 - Polymerization temperature: 100°C — Examples 17-23 - Polymerization time: 9 h — Examples 17-23

Relevant Experimental Evidence - Example 1: Standard solution NBR synthesis from acrylonitrile and butadiene at 100°C and 3 bar with free-radical initiator/modifier system. - Example 6: Terpolymerization with hydroxyethyl acrylate (HEA) as termonomer; reaction stopped after 6 h at 39% conversion. - Example 7: Low-molecular-weight NBR was intentionally prepared by increasing regulator amount to 2.22 phm. - Example 8: RAFT macromer NBR #7 (Mn = 12,000 g/mol) was chain-extended with styrene at 100°C. - Examples 17-23: NBR copolymerizations with

HDODA and/or HEA, including post-dosing of initiator to sustain conversion.

Technical Relevance This is one of the most directly relevant patents because it explicitly targets low-molecular-weight NBR through solution polymerization and regulator adjustment. It provides concrete process conditions for controlling molecular weight and isolating the polymer by precipitation.

Patent 2

- Patent Number: US10167348B2
- Patent Title: Solution polymers formed from methylene malonate monomers, polymerization, and ...
- Assignee: Not disclosed
- Jurisdiction: US
- Publication Year: 2019
- Legal Status: Unknown
- Polymer Type: Not disclosed
- Relevance: Automatically extracted candidate

Polymerization / Synthesis Method - Synthesis method: This patent discloses solution anionic polymerization of 1,1-disubstituted alkene compounds using weak-base or organolithium activation, with molecular weight controlled by monomer-to-activator ratio, temperature, solvent, and monomer feed strategy. Although not an NBR patent, it is highly relevant as a controlled solution-polymerization framework for low-viscosity, low-MW polymer synthesis and scale-up. - Monomer to activator ratio: about 1000:1 — Example H-1 - Reaction time: about 1 h — Example H-1 - Reaction temperature: about 23°C — Example H-1 - Cold methanol precipitation temperature: 0°C — Example H-1 - PDI: about 1.43 — Example H-1 - First peak molecular weight: about 2000 — Example H-1 - Second peak molecular weight: about 60,000 — Example H-1 - Number average molecular weight: about 40,000 — Example H-2 - Reaction temperature: about -78°C — Example N-1 - Monomer to activator ratio: about 1000:1 — Example N-1 - Reaction time: about 20 min — Example N-1 - Mn: 12,750 to 41,280 daltons — Example N-1 - Monomer to activator ratio: about 100:1 — Example N-3 - Total monomer to activator ratio: about 1000:1 — Example N-2 - Mn: 48,000 to 130,000 daltons — Example N-2 - Total monomer to activator ratio: about 1000:1 — Example R-1 - Reaction temperature: about 23°C — Example R-1 - Mn: about 7,104 daltons — Example R-1 - Mw: about 16,343 daltons — Example R-1 - PDI: about 2.3 — Example R-1 - HM3:DEMM ratio: 75:25, 50:50, 25:75 — Examples R-2 to R-4 - Reaction time: about 1 h — Examples R-2 to R-4 - Reaction temperature: about 23°C — Examples R-2 to R-4 - Monomer to activator ratio: about 100:1 — Example P-1 - Monomer to activator ratio: about 1000:1 — Example P-2 - Number average molecular weight: about 6,700 daltons — Example P-1 - Number average molecular weight: about 1,451,800 daltons — Example P-2 - Weight average molecular weight: about 2,239,300 daltons — Example P-2 - Reaction temperature: about 23°C — Example X-4 - Reaction time: 1 h — Example X-4 - Solvent: tetrahydrofuran — Examples H, N, R, X

Relevant Experimental Evidence - Example H-1: Demonstrated solution polymerization of a methylene malonate monomer with controlled molecular weight and narrow PDI. - Example N-1: Showed low-temperature anionic polymerization at -78°C with M_n increasing from 12,750 to 41,280 daltons over 20 min. - Example R-1: Produced a random copolymer with M_n 7,104 daltons and M_w 16,343 daltons at 23°C . - Example B-1: Built an A-B-A-B block copolymer with near-quantitative conversion and M_n increasing to 67,400 daltons. - Example X-4: Demonstrated scale-up to 450 g polymer while maintaining $\sim 100\%$ conversion and M_n around 528,400 daltons.

Technical Relevance While not NBR-specific, this patent is relevant because it provides a detailed solution-polymerization playbook for controlling molecular weight, viscosity, and architecture. Those same control concepts are transferable to low-MW NBR synthesis and process design.

Patent 3

- Patent Number: US8470957B2
- Patent Title: Method for producing bischloroformate compound, polycarbonate oligomer having ...
- Assignee: Not disclosed
- Jurisdiction: US
- Publication Year: 2013
- Legal Status: Unknown
- Polymer Type: Not disclosed
- Relevance: Automatically extracted candidate

Polymerization / Synthesis Method - Synthesis method: This patent discloses production of bischloroformate compounds from dihydric phenols and phosgene in dichloromethane using tertiary amines, with emphasis on producing monomer-like low-oligomer products. It is not an NBR synthesis patent, but it is relevant as a process-control reference for low-degree-of-polymerization intermediates and continuous/microreactor handling. - TEA: 59.8 g (0.591 mol) — Example 1-1 - MDC: 100 mL — Example 1-1 - 4,4'-biphenol: 50.0 g (0.269 mol) — Example 1-1 - Phosgene: 80.0 g (0.809 mol) — Example 1-1 - Reaction temperature: 13 to 16°C — Example 1-1 - Reaction time: 3 h 6 min — Example 1-1 - Reaction temperature: 14 to 16°C — Example 1-1 - Reaction time: 1 h 38 min — Example 1-1 - BPZ: 35 g (0.13 mol) — Example 1-2 - Phosgene: 38.7 g (0.39 mol) — Example 1-2 - Reaction temperature: 5 to 17°C — Example 1-2 - Reaction time: 3 h — Example 1-2 - Triphosgene: 38.7 g (0.13 mol) — Example 1-3 - Reaction temperature: 3 to 5°C — Example 1-3 - Reaction time: 26 min — Example 1-3 - Reaction temperature: 11 to 18°C — Example 1-3 - Bis-OCF: 230 g (1.01 mol) — Example 1-4 - Phosgene: 223 g (2.25 mol) — Example 1-4 - Reaction temperature: 15.5 to 19°C — Example 1-4 - Reaction time: 3 h 3 min — Example 1-4 - Material solution flow rate: 50 mL/min — Example 4-1 - Phosgene solution flow rate: 37.2 mL/min — Example 4-1 - Linear velocity: 5.8 mL/s — Example 4-1 - Theoretical phosgene equivalent: 1.5 equivalent weight — Example 4-1 - Microreactor temperature: 0°C — Example 5-1 - Material solution flow rate: 1.5 mL/min — Example 5-1 - Phosgene solution flow rate: 1.08 mL/min — Example 5-1 - Average number of repeating units (n): 1.99 or less —

Abstract/claims

Relevant Experimental Evidence - Example 1-1: Prepared a bischloroformate-containing solution from biphenol and phosgene in MDC with TEA. - Example 1-3: Showed triphosgene-based bischloroformate formation at low temperature. - Example 4-1: Demonstrated continuous/microreactor production with controlled flow rates and theoretical phosgene equivalent of 1.5. - Example 5-1: Showed microreactor operation at 0°C with low feed rates. - Experimental Example 3-1: Produced a polycarbonate oligomer with reduced viscosity [η_{sp}/c] of 0.94.

Technical Relevance This patent is only indirectly relevant to low-MW NBR, but it is useful as a low-oligomer/controlled-reactivity process reference. It highlights how feed control and microreactor operation can be used to target monomer-like or oligomeric products.

Patent 4

- Patent Number: US9346908B1
- Patent Title: Olefin-metathesis catalysts and method for preparing low-molecular-weight ...
- Assignee: Not disclosed
- Jurisdiction: US
- Publication Year: 2016
- Legal Status: Unknown
- Polymer Type: Not disclosed
- Relevance: Automatically extracted candidate

Polymerization / Synthesis Method - Synthesis method: This patent discloses reduction of NBR molecular weight by olefin-metathesis in chlorobenzene using a ruthenium-based catalyst system, including loaded catalysts on magnetic carriers for easy recovery. The NBR solution is reacted with a co-olefin such as 1-hexene at room temperature, then the catalyst is magnetically separated and reused. - NBR acrylonitrile content: 38 mol % — Comparative Example 1 / Examples 7-9 - NBR solid content: 6 wt % — Comparative Example 1 / Examples 7-9 - 1-hexene co-olefin: 2 wt % relative to NBR solid content — Comparative Example 1 / Examples 7-9 - Loaded catalyst amount: 5 wt % relative to NBR solid content — Examples 7-9 - Reaction time: 2 h — Comparative Example 1 / Examples 7-9 - Reaction temperature: room temperature — Comparative Example 1 / Examples 7-9 - Mw before reaction: 126,049 — Comparative Example 1 / Examples 7-9 - Mw after reaction: 45,925 to 72,893 — Example 7 - Mw after reaction: 41,583 to 64,163 — Example 8 - Mw after reaction: 39,054 to 60,992 — Example 9 - PDI before reaction: 1.98 — Comparative Example 1 / Examples 7-9 - PDI after reaction: 1.59 to 1.82 — Examples 7-9 - Degradation rate: 63.6% to 69.0% in first round — Examples 7-9 - Catalyst recovery/reuse: repeated five times — Comparative Example 1 / Examples 7-9

Relevant Experimental Evidence - Comparative Example 1: Commercial loaded catalyst reduced Mw from 126,049 to 66,190 after one round, but degradation rate was below 50% after repeated reuse. - Example 7: Fe-StVP1-GII catalyst reduced Mw to 45,925 after round 1 and maintained activity through five reuse cycles. - Example 8: Fe-StVP2-GII catalyst gave Mw 41,583

after round 1 and 64,163 after round 5. - Example 9: Fe-StVP2-Z1 catalyst gave the strongest initial degradation, with 69.0% in round 1 and 51.6% after round 5. - All examples: Catalyst was recovered by magnetic separation and reused.

Technical Relevance This is one of the most directly relevant patents for low-MW NBR because it explicitly converts high-MW NBR into lower-MW NBR by metathesis. It also addresses a key industrial issue: catalyst recovery from viscous NBR solutions.

Patent 5

- Patent Number: US11905379B2
- Patent Title: Associative polymers and related compositions, methods and systems
- Assignee: Not disclosed
- Jurisdiction: US
- Publication Year: 2024
- Legal Status: Unknown
- Polymer Type: Not disclosed
- Relevance: Automatically extracted candidate

Polymerization / Synthesis Method - Synthesis method: This patent discloses associative polymers based on non-polar backbones, especially telechelic 1,4-polybutadiene, prepared by ROMP and chain-transfer-agent strategies and then end-functionalized to create self-associating or donor-acceptor systems. Although not an NBR patent, it is relevant to low-viscosity polymer design and to the use of controlled molecular weight and end-group chemistry to tune rheology. - CTA amount: 6.7 mg — Example 3 - CTA solution volume: 0.5 mL deoxygenated DCM — Example 3 - Grubbs II catalyst solution: 0.13 mL of 1 mg/mL DCM solution — Example 3 - COD: 0.03 mL ($\approx$ 50 eq. w.r.t. CTA) — Example 3 - Reaction temperature: 40°C — Example 3 - Reaction time: 33 min — Example 3 - Second COD charge: 5.6 mL ($\approx$ 10,000 eq.) in 12 mL DCM — Example 3 - Mw: 430,000 g/mol — Example 3 - PDI: 1.46 — Example 3 - TFA volume: 1.25 mL — Example 4 - TFA hydrolysis: overnight at room temperature — Example 4 - Azidotrimethylsilane: 0.73 g ($\approx$ 1200 eq.) — Example 5 - Tetrabutylammonium fluoride: 1.57 g ($\approx$ 1200 eq.) — Example 5 - Reaction temperature: 60°C — Example 5 - 3-Dimethylamino-1-propyne: 0.23 g ($\approx$ 1200 eq.) — Example 5 - PMDETA: 0.02 g ($\approx$ 50 eq.) — Example 5 - Copper(I) bromide: 0.016 g ($\approx$ 50 eq.) — Example 5 - Reaction temperature: 50°C — Example 5 - Apparent Mw increase: 20% after TFA hydrolysis — Example 6 - 1 wt % Jet-A solutions — Examples 6-9 - 0.5% di-TA PB in Jet-A — Example 8 - 0.3 wt % Jet-A solution of 430K di-TA PB — Example 9 - Mw of telechelic PB: 630,000 g/mol, 540,000 g/mol, 670,000 g/mol — Examples 13A, 13B - Polymer backbone lengths: 76,000, 230,000, 430,000 g/mol — Example 14

Relevant Experimental Evidence - Example 3: Built a 430K di-TE polybutadiene prepolymer by ROMP using octa-functional CTA and Grubbs II catalyst. - Example 5: Converted the prepolymer into tertiary-amine-terminated telechelic PB through azidation and click-type end functionalization. - Example 6: Showed that TFA deprotection increased apparent Mw by 20% due to self-association. - Example 7: Demonstrated donor/acceptor end-group association in 1 wt %

Jet-A solutions. - Example 8: Showed that 0.5% associative polymer remained soluble in Jet-A at -30°C for months.

Technical Relevance This patent is not NBR-specific, but it is relevant because it demonstrates how controlled molecular weight and end-group chemistry can be used to tune viscosity and solution behavior. Those design principles are useful when considering low-MW NBR or NBR-derived additives.

Patent 6

- Patent Number: US10731046B2
- Patent Title: Biopolymer-based inks and use thereof
- Assignee: Not disclosed
- Jurisdiction: US
- Publication Year: 2020
- Legal Status: Unknown
- Polymer Type: Not disclosed
- Relevance: Automatically extracted candidate

Polymerization / Synthesis Method - Synthesis method: This patent discloses aqueous biopolymer inks, including low-molecular-weight structural protein formulations, for inkjet printing and related deposition methods. It is not an NBR synthesis patent, but it provides a process framework for low-viscosity printable polymer solutions and controlled droplet formation. - Molecular weight: about 3.5 kD to about 350 kD — Claim 1 - Keratin concentration: about 0.1 wt/vol % to about 10 wt/vol % — Claim 1 - Droplet volume: about 0.1 pL to about 5 nL — Claim 1 - Viscosity: about 1 cP to about 20 cP — Claim 1 - Surface tension: about 15 dynes/cm to about 50 dynes/cm — Claim 1 - pH: between 5-9 — Claim 2 - Viscosity-modifying agent: about 5-30 wt % — Claim 4 - Surfactant: about 0.1-10 wt % — Claim 5 - Low molecular weight structural protein concentration: about 0.1-10.0% — Description - Droplet diameter: about 0.1-250 μm — Description - Resolution: about 50-20,000 dpi — Description - Room temperature: about 18-26 $^{\circ}\text{C}$ — Description

Relevant Experimental Evidence - Example 1: Demonstrated silk fibroin ink feasibility for inkjet printing. - Example 2: Prepared keratin ink from wool and concentrated it to 7 wt%. - Example 3: Discussed silk fibroin ink printing and biocompatibility context.

Technical Relevance This patent is only indirectly relevant to low-MW NBR, but it is useful as a benchmark for low-viscosity polymer solution design and printability. It highlights the importance of molecular weight and viscosity windows for processable polymer formulations.

Patent 7

- Patent Number: US10336837B2
- Patent Title: Process for producing olefin polymer and olefin polymer

- Assignee: Not disclosed
- Jurisdiction: US
- Publication Year: 2019
- Legal Status: Unknown
- Polymer Type: Not disclosed
- Relevance: Automatically extracted candidate

Polymerization / Synthesis Method - Synthesis method: This patent discloses olefin polymerization using metallocene-type catalysts under pressure, including 1-butene homopolymerization and copolymerization with propylene or higher alpha-olefins. It is not an NBR patent, but it is relevant as a controlled polymerization reference showing how catalyst choice, hydrogen, temperature, and comonomer feed affect molecular weight and viscosity-related properties. - 1-butene supply rate: 90 L/h — Example 1A - Triisobutylaluminum: 0.25 mmol — Example 1A - Catalyst B: 10 μ mol — Example 1A - MMAO-3A: 5.0 mmol (Al atoms) — Example 1A - Polymerization temperature: 25°C — Example 1A - Polymerization time: 45 min — Example 1A - Hydrogen: 0.25 L/h — Example 2A - Catalyst C: 0.60 μ mol — Example 2A - MMAO-3A: 0.30 mmol — Example 2A - Polymerization temperature: 25°C — Example 2A - Polymerization time: 45 min — Example 2A - Heptane: 500 mL — Example 5A - 1-butene: 100 g — Example 5A - Autoclave pressure: 0.50 MPaG — Example 5A - Polymerization temperature: 70°C — Example 5A - Polymerization time: 20 min — Example 5A - 1-butene: 180 g — Example 7A - Polymerization temperature: 50°C — Example 7A - Polymerization temperature: 60°C — Example 8A - Hydrogen: 0.06 NL — Example 9A - Polymerization temperature: 60°C — Example 9A - Catalyst D: 4.3 μ mol — Table 3-1 - Catalyst D: 0.23 μ mol — Table 3-1 - Catalyst D: 0.46 μ mol — Table 3-1 - Catalyst b: 3.9 μ mol — Table 8 - Catalyst a: 3.8 μ mol — Table 8 - Mw: 243×10^4 — Example 2A - Mw/Mn: 2.75 — Example 2A - Mw: 163×10^4 — Example 4A - Mw/Mn: 2.53 — Example 4A

Relevant Experimental Evidence - Example 1A: Produced 1-butene polymer at 25°C with catalyst B and MMAO-3A. - Example 2A: Added hydrogen during polymerization and obtained higher Mw polymer than Example 1A. - Example 5A: Ran 1-butene homopolymerization at 70°C and 0.50 MPaG. - Example 7A: Increased 1-butene charge to 180 g and changed temperature to 50°C. - Example 9A: Added 0.06 NL hydrogen and polymerized at 60°C.

Technical Relevance This patent is not NBR-specific, but it is relevant as a controlled olefin-polymerization reference for how catalyst system, hydrogen, and temperature influence molecular weight and rheology. Those principles are useful when designing low-MW polymerization strategies.

Patent 8

- Patent Number: US9988480B2
- Patent Title: Pigment dispersant, production method for pigment dispersant, and pigment ...
- Assignee: Not disclosed
- Jurisdiction: US

- Publication Year: 2018
- Legal Status: Unknown
- Polymer Type: Not disclosed
- Relevance: Automatically extracted candidate

Polymerization / Synthesis Method - Synthesis method: This patent discloses living radical polymerization to make graft copolymer pigment dispersants using iodine-initiated systems, acidic or basic methacrylates, and methacrylate-based macromonomers. It is not an NBR patent, but it is relevant because it demonstrates controlled radical polymerization, low-MW side chains, and viscosity/stability tuning in solution. - Iodine: 3.1 parts (0.0122 mol) — Synthesis Example 1 - V-70: 14.8 parts (0.0480 mol) — Synthesis Example 1 - PGMAc: 426.8 parts — Synthesis Example 1 - DPM: 0.5 parts — Synthesis Example 1 - BzMA: 33.3 parts (0.189 mol) — Synthesis Example 1 - PA: 83.5 parts (0.300 mol) — Synthesis Example 1 - PP500: 175 parts (0.299 mol) — Synthesis Example 1 - Monomer/initiating compound ratio: 32.3/1 — Synthesis Example 1 - Polymerization temperature: 40°C — Synthesis Example 1 - Polymerization time: 9 h — Synthesis Example 1 - Mn: 8500 — Synthesis Example 2 - PDI: 1.30 — Synthesis Example 2 - Solid content: 40.5% — Synthesis Example 2 - Acid value: 22.2 mgKOH/g — Synthesis Example 2 - Monomer/initiating compound ratio: 25.4/1 — Synthesis Example 3 - Mn: 17500 — Synthesis Example 3 - PDI: 1.50 — Synthesis Example 3 - Acid value: 39.8 mgKOH/g — Synthesis Example 3 - CLMA molecular weight: 3000 — Synthesis Example 4 - Mn: 22600 — Synthesis Example 4 - PDI: 1.45 — Synthesis Example 4 - Acid value: 30.7 mgKOH/g — Synthesis Example 4 - DBE solvent: 64 parts — Synthesis Example 5 - MAA: 12.9 parts (0.150 mol) — Synthesis Example 5 - MMA: 7.5 parts (0.075 mol) — Synthesis Example 5 - Mn: 14900 — Synthesis Example 5 - PDI: 1.32 — Synthesis Example 5 - DMAEMA: 47.1 parts (0.300 mol) — Synthesis Example 6 - PP800: 240 parts (0.271 mol) — Synthesis Example 6 - Mn: 12800 — Synthesis Example 6 - PDI: 1.33 — Synthesis Example 6 - Amine value: 54.6 mgKOH/g — Synthesis Example 6 - Mn: 16000 — Synthesis Example 7 - PDI: 1.44 — Synthesis Example 7 - Amine value: 43.6 mgKOH/g — Synthesis Example 7 - CLMA: 180 parts (0.060 mol) — Synthesis Example 8 - Mn: 10200 — Synthesis Example 8 - PDI: 1.43 — Synthesis Example 8 - Amine value: 67.9 mgKOH/g — Synthesis Example 8 - Monomer/initiating compound ratio: 38/1 — Synthesis Example 9 - Mn: 17300 — Synthesis Example 9 - PDI: 1.41 — Synthesis Example 9 - Amine value: 40.0 mgKOH/g — Synthesis Example 9 - Polymerization temperature: 45°C — Synthesis Example 10 - Polymerization time: 8 h — Synthesis Example 10 - Mn: 4900 — Synthesis Example 10 - PDI: 1.22 — Synthesis Example 10 - Polymerization temperature: 45°C — Synthesis Example 11 - Polymerization time: 8 h — Synthesis Example 11 - Mn: 5900 — Synthesis Example 11 - PDI: 1.21 — Synthesis Example 11

Relevant Experimental Evidence - Synthesis Example 1: Prepared acidic pigment dispersant-1 by iodine-initiated living radical polymerization at 40°C. - Synthesis Example 2: Achieved Mn 8500 and PDI 1.30 with near-complete conversion. - Synthesis Example 4: Used CLMA macromonomer to obtain Mn 22600 and PDI 1.45. - Synthesis Example 6: Prepared basic pigment dispersant-1 with Mn 12800 and amine value 54.6 mgKOH/g. - Synthesis Example 10: Demonstrated a two-step method giving Mn 4900 and PDI 1.22.

Technical Relevance This patent is indirectly relevant because it shows how living radical polymerization can precisely control molecular weight and architecture in solution. Those same control concepts are useful for designing low-viscosity or low-MW NBR-related polymer systems.

Patent 9

- Patent Number: US10465017B2
- Patent Title: Bulk polymerization of silicone-containing copolymers
- Assignee: Not disclosed
- Jurisdiction: US
- Publication Year: 2019
- Legal Status: Unknown
- Polymer Type: Not disclosed
- Relevance: Automatically extracted candidate

Polymerization / Synthesis Method - Synthesis method: This patent discloses essentially adiabatic bulk polymerization of silicone-containing copolymers using thermal initiators, chain transfer agents, and staged addition of initiators/components. It is not an NBR patent, but it is relevant as a process-control reference for managing exotherm, molecular weight, and viscosity in bulk radical polymerization. - IOA: 83 parts by weight — Comparative Example C-1 - AA: 7 parts by weight — Comparative Example C-1 - SiMac: 10 parts by weight — Comparative Example C-1 - Initiator-5: 0.3 parts by weight — Comparative Example C-1 - EA: 150 parts by weight — Comparative Example C-1 - Reaction temperature: 55°C — Comparative Example C-1 - Reaction time: 24 h — Comparative Example C-1 - Dry coating thickness: 25 micrometers — Comparative Example C-1 - Reaction temperature: 60°C — Examples 1-4 - Typical reaction peak temperature: 150°C — Examples 1-4 - Second-step peak temperature: 160°C — Examples 1-4 - UV dose: 600 mJ/cm² — Examples 1-4 - IV: 0.55 to 0.29 dl/g — Examples 1-4 - 1-butene polymerization temperature: 70°C — Example 5A - Polymerization time: 20 min — Example 5A - 1-butene polymerization temperature: 40°C — Example 6A - 1-butene amount: 180 g — Example 7A - 1-butene polymerization temperature: 50°C — Example 7A - 1-butene polymerization temperature: 60°C — Example 8A - Hydrogen: 0.06 NL — Example 9A - Polymerization temperature: 60°C — Example 9A - Extrusion temperatures: 110, 115, 120, 140, 140°C — Example 10 - Extrusion pressure: 2,960 to 3,450 kPa — Example 10 - Extrusion rate: 90 RPM — Example 10 - Typical reaction peak temperature: 170°C — Example 11 - Typical reaction peak temperature: 190°C — Example 11

Relevant Experimental Evidence - Examples 1-4: Demonstrated essentially adiabatic two-step polymerization with peak temperatures around 150-160°C and IV values from 0.55 to 0.29 dl/g. - Example 10: Showed extrusion of a polymer blend at 110-140°C with transparent release film formation. - Example 11: Produced a polymer with IV 0.25 dl/g using staged initiator addition and adiabatic polymerization. - Example 12: Converted the polymer into pellets and then a two-layer film. - Example 13: Showed peel/adhesion performance after 10 days at 65°C.

Technical Relevance This patent is not NBR-specific, but it is relevant because it demonstrates how staged radical polymerization and thermal management can be used to control molecular weight and processability. Those principles are transferable to low-MW NBR process development.

Patent 10

- Patent Number: US20200354606A1
- Patent Title: Composition, methods and devices useful for manufacturing of implantable ...
- Assignee: Not disclosed
- Jurisdiction: US
- Publication Year: 2020
- Legal Status: Unknown
- Polymer Type: Not disclosed
- Relevance: Automatically extracted candidate

Polymerization / Synthesis Method - Synthesis method: This patent discloses aqueous dispersible biodegradable compositions formed from a polyol/diacid ester, especially glycerol-sebacic acid ester compounds, within a hydrated polypeptide matrix. It is not an NBR patent and does not disclose NBR synthesis, but it is relevant as a process/formulation reference for low-viscosity, printable, and extrudable polymer compositions. - Polymeric glycerol-sebacic acid ester compound molecular weight: in excess of 10,000 — Claim 4 - Oligomeric glycerol-sebacic acid ester compound molecular weight: less than 10,000 — Claim 5 - Polypeptide hydrogel formation: solid polypeptide in water — Claim 1 - Addition of glycerol-sebacic acid ester compound while molten or solid — Claims 2-3 - Collagen or gelatin: selected solid polypeptide — Claim 8

Relevant Experimental Evidence - Claimed method: Mixed solid polypeptide in water to form a hydrogel and added glycerol-sebacic acid ester compound. - Claimed composition: Aqueous dispersion of glycerol-sebacic acid ester compound in polypeptide hydrogel. - Claimed article formation: Extrusion-based 3D printing and fiber formation were described.

Technical Relevance This patent is only indirectly relevant to low-MW NBR, but it is useful as a formulation/process reference for dispersible, extrudable polymer systems. It highlights how molecular-weight class and matrix composition can be used to tune handling and printability.

3. CROSS-PATENT COMPARISON & SYNTHESIS TRENDS

- EP2478021B1 and US9346908B1 are the most directly relevant NBR patents: EP2478021B1 controls NBR molecular weight during solution polymerization with a regulator, while US9346908B1 lowers the molecular weight of preformed NBR by olefin-metathesis

degradation.

- EP2478021B1 uses acrylonitrile/1,3-butadiene solution polymerization in DMAc with free-radical initiator plus DoPAT regulator; US9346908B1 uses chlorobenzene NBR solutions with 1-hexene and loaded ruthenium metathesis catalysts.
- EP2478021B1 explicitly reports low-MW NBR by increasing regulator amount to 2.22 phm and gives $M_n = 12,000$ g/mol for a RAFT macromer; US9346908B1 reports M_w reduction from 126,049 to 39,054-45,925 after one metathesis treatment.
- EP2478021B1 relies on precipitation in ethanol and vacuum drying for isolation; US9346908B1 isolates product by methanol precipitation and recovers catalyst magnetically.
- US10167348B2, US11905379B2, US9988480B2, and US10465017B2 are not NBR-specific but collectively reinforce the importance of solution polymerization, chain-transfer control, living radical control, and staged thermal management for molecular-weight tuning.
- US10731046B2 and US20200354606A1 are formulation/process references rather than NBR synthesis disclosures; both emphasize low-viscosity or printable compositions, which is relevant to downstream handling of low-MW NBR-like materials.
- US8470957B2 is the least directly related to NBR chemistry, but it contributes a low-oligomer, feed-controlled microreactor strategy that is conceptually aligned with controlled low-MW product formation.

4. CONCLUSION

The evidence supports two dominant routes to low molecular weight NBR: controlled solution polymerization with a regulator/chain-transfer strategy, and post-polymer molecular-weight reduction by olefin metathesis. EP2478021B1 is the clearest synthesis patent for low-MW NBR formation, showing that increasing regulator loading in DMAc solution at 100°C and 3 bar butadiene can deliberately suppress molecular weight. US9346908B1 provides a complementary degradation route, where 6 wt% NBR solutions treated with 2 wt% 1-hexene and 5 wt% magnetic loaded ruthenium catalyst at room temperature for 2 h yield substantially lower M_w and allow catalyst recovery. For practical low-MW NBR synthesis, the strongest evidence favors solvent-based control, oxygen exclusion, moderate temperature, and either elevated regulator loading during polymerization or metathesis-based chain scission after polymerization. The broader patent set reinforces that precise molecular-weight control, low-viscosity formulation, and recoverable catalyst/process design are the key enabling parameters.

5. REFERENCES

- EP2478021B1 | Nitrile rubbers and production of same in organic solvents | Not disclosed | EP | 2015 | <https://patents.google.com/patent/EP2478021B1/en>
- US10167348B2 | Solution polymers formed from methylene malonate monomers, polymerization, and ... | Not disclosed | US | 2019 | <https://patents.google.com/patent/US10167348B2/en>

- US8470957B2 | Method for producing bischloroformate compound, polycarbonate oligomer having ... | Not disclosed | US | 2013 | <https://patents.google.com/patent/US8470957B2/en>
- US9346908B1 | Olefin-metathesis catalysts and method for preparing low-molecular-weight ... | Not disclosed | US | 2016 | <https://patents.google.com/patent/US9346908B1/en>
- US11905379B2 | Associative polymers and related compositions, methods and systems | Not disclosed | US | 2024 | <https://patents.google.com/patent/US11905379B2/en>
- US10731046B2 | Biopolymer-based inks and use thereof | Not disclosed | US | 2020 | <https://patents.google.com/patent/US10731046B2/en>
- US10336837B2 | Process for producing olefin polymer and olefin polymer | Not disclosed | US | 2019 | <https://patents.google.com/patent/US10336837B2/en>
- US9988480B2 | Pigment dispersant, production method for pigment dispersant, and pigment ... | Not disclosed | US | 2018 | <https://patents.google.com/patent/US9988480B2/en>
- US10465017B2 | Bulk polymerization of silicone-containing copolymers | Not disclosed | US | 2019 | <https://patents.google.com/patent/US10465017B2/en>
- US20200354606A1 | Composition, methods and devices useful for manufacturing of implantable ... | Not disclosed | US | 2020 | <https://patents.google.com/patent/US20200354606A1/en>